Valerio Dionisi

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GitHub Rome (It), 04/07/1997

Research Interests

Macroeconomics, *Production Networks*, *Structural Macro-Labour*, *Fiscal and Monetary Policy*

Education

2024 – 2025 ***Visiting Ph.D. Student*** — University of Oxford

Activities: Production networks, macro-labour, fiscal policy
Advisor: Prof. **Francesco Zanetti**

2023 – 2024 ***Visiting Ph.D. Student*** — Universitat Pompeu Fabra

Activities: Numerical techniques for quantitative macroeconomics, production networks
Advisor: Prof. **Davide Debortoli**

2021 – 2026 **Ph.D.** in Economics, Statistics and Data Science (cum laude) — University of Milano-Bicocca

Thesis: "Macroeconomic Perspectives on Sectoral Composition: Networks and Inequality"
Committee: Basile Grassi (Bocconi), Alessio Moro (UniCa), Francesco Zanetti (Oxford), Enzo Dia (UniMiB)
Supervisor: Prof. **Andrea Colciago**

2019 – 2021 **Ms.C.** in European Economy and Business Law (cum laude) — Tor Vergata University

Thesis: "Evidence of Debt Consolidation Dynamics for Europe and Italy"
Supervisor: Prof. **Barbara Annicchiarico**, *Co-supervisor*: Prof. **Leonardo Becchetti**

2016 – 2019 **B.A.** in Political Sciences — Roma Tre University

Thesis: "Instrument-Target Alignment and the Coordination of Economic Policies"
Supervisor: Prof. **Gian Cesare Romagnoli**

Further Education

2022 ***Advanced Computational Macroeconomics*** — London School of Economics

Research Papers

Job Market Paper ***The Horizontal Geometry of Production Networks*** — go to the paper

Modern economies are organized as dense production networks, yet comovement is typically understood through observed Input-Output linkages and vertical cascade effects. This paper argues that this view is incomplete. Comovement fundamentally depends on horizontal complementarities: the extent to which sectors share upstream suppliers or downstream buyers. Such network similarities in demand and supply induce simultaneous responses to idiosyncratic shocks, shaping the sign of comovement. Regarding aggregate fluctuations, horizontal transmission generates systematic comovement that prevents shocks from averaging out, without relying on cascading amplification. The framework developed highlights this novel diffusion channel as a complement to vertical propagation, offering a unified view of comovement rooted in the horizontal geometry of common economic interactions rather than network trade intensity. Using Input-Output data, I show that sectoral employment responses vary systematically with horizontal distances, with nearby sectors exhibiting dampened or opposing responses, and distant ones comoving positively.

Working Papers ***Networks, Real Rigidities, and Monetary Policy*** DRAFT UPON REQUEST

with Colciago A. (DNB, UniMiB), Siena D. (PoliMi, Bocconi), and Zago R. (BdF, CdF, ESCP)

Over the past decades, the slope of the Phillips Curve has exhibited substantial variation across countries and sectors, with a marked flattening prior to the pandemic. This paper studies how the topology of domestic and global production networks shapes inflation dynamics. Using country-industry data on producer prices and output, we document systematic heterogeneity in Phillips-curve slopes: downstream sectors and industries with strong backward integration display flatter inflation responses, while upstream sectors and industries with strong forward linkages exhibit steeper dynamics. We rationalize these asymmetries with a network-based DSGE model where the NKPC slope reflects the interaction between real rigidities (strategic complementarities and endogenous markups), network propagation of nominal rigidities, and global slack transmission through imported intermediates. The results imply that the aggregate Phillips Curve slope is an endogenous object that shifts with production network structure and global integration.

Industrial Structure, Technological Change, and U.S. Wage Inequality *submitted*, GO TO THE PAPER

This paper examines the determinants of U.S. wage inequality between industries. It distinguishes two channels of Skill-Biased Technological Change: a quantity effect, mirroring variations in the distribution of capital and labour inputs, and a structural effect, reflecting differences in substitution elasticities across factors of production. Counterfactual exercises show that evolving technological heterogeneity in substitution patterns dominates wage dispersion. A quantitative decomposition reveals that structural forces, mostly reflecting substitutability of routine with non-routine labour, account for 94% of observed wage variance, exacerbated by stronger sorting and segregation effects. Upon neutralising structural differences, the quantity channel reckons merely 6-15%. Indeed, structural heterogeneity, rather than mere changes in factor quantities, is the dominant force behind U.S. wage inequality.

Work in Progress ***Vertically-Integrated Networks and the Consumption Channel***

Job Vacancies through COVID-19 Lockdowns in the Euro Area *with Bertucci A. (UniPv)*

Where to Sit in the Chain: Microfoundations and Micro-to-Macro Implications of Global Value-Chain Positioning *with Siena D. (PoliMi, Bocconi)*

Teaching

2024/2025-*still*	***International Macroeconomics*** *(Grad), TA Lecturer*	Polytechnic University of Milan
2024/2025-*still*	***Macroeconomics*** *(UndGrad), TA Lecturer*	Polytechnic University of Milan
2023/2024-*still*	***Computational Macroeconomics*** *(Grad), TA Lecturer*	University of Milano-Bicocca
2023/2024-*still*	***Dynamic Asset Pricing*** *(Grad), TA*	University of Milano-Bicocca
2023/2024-*still*	***Economia dei Contratti Finanziari*** *(Grad), TA*	University of Milano-Bicocca
2023/2024-*still*	***Macroeconomics*** *(UndGrad), TA*	University of Milano-Bicocca

Skills

Languages:	Italian (native), English (fluent), French and Spanish (intermediate)
Programming:	MatLab, Dynare, Stata (advanced); Julia, R (proficient); LaTeX (advanced)
Microsoft Office:	Word, Excel, PowerPoint, Outlook

Certificates

2021	Introduction to Linear Algebra	MatLab
2021	Introduction to Statistical Methods	MatLab
2021	Fundamentals	MatLab
2021	OnRamp	MatLab

2021 Optimization OnRamp — MatLab

Conferences and Seminars, *including scheduled*

2026 4th Milan Ph.D. Economics Workshop, *Organizer*; 25th Research in International Economics and Finance Workshop (***scheduled***), International Association for Applied Econometrics (IAAE) Annual Conference (***scheduled***), Irish Economic Association Annual Conference (***scheduled***), 1st Piraeus International Economic Conference (***scheduled***), Tor Vergata University (***scheduled***), Roma 3 University, Catholic University of Milan, University of Milano-Bicocca, *Presenter*

2025 3rd Milan Ph.D. Economics Workshop, *Organizer*; University of Oxford, Naples School of Economics: 4th PhD and Post-Doctoral Workshop, University of Naples "Federico II", "Sapienza" University of Rome, Catholic University of Milan, University of Milano-Bicocca, *Presenter*

2024 2nd Milan Ph.D. Economics Workshop, *Organizer*, University of Pavia, Catholic University of Milan, University of Milano-Bicocca *Presenter*

2023 1st Milan Ph.D. Economics Workshop, University of Milan, University of Milano-Bicocca, Catholic University of Milan, *Presenter*

References

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